Department of Electronics and Electrical Engineering
Indian Institute of Technology Guwahati

EE101 - Basic Electronics

Date: Aug 19, 2023 **Pre-Tutorial Assignment 2** **Marks: 2**

1. Figure 1 shows an electric circuit with independent and dependent current sources. The numerical values of the circuit elements are as follows: $I_1 = 5A$, $I_2 = k \times i_i = 5 \times i_1$, $R_1 = 1\Omega$, $R_2 = 2\Omega$, and $R_3 = 5\Omega$. Evaluate the branch currents i_1, i_2 and i_3 using nodal analysis. The nodes **1**, 2, and reference node **R** are marked in the figure.

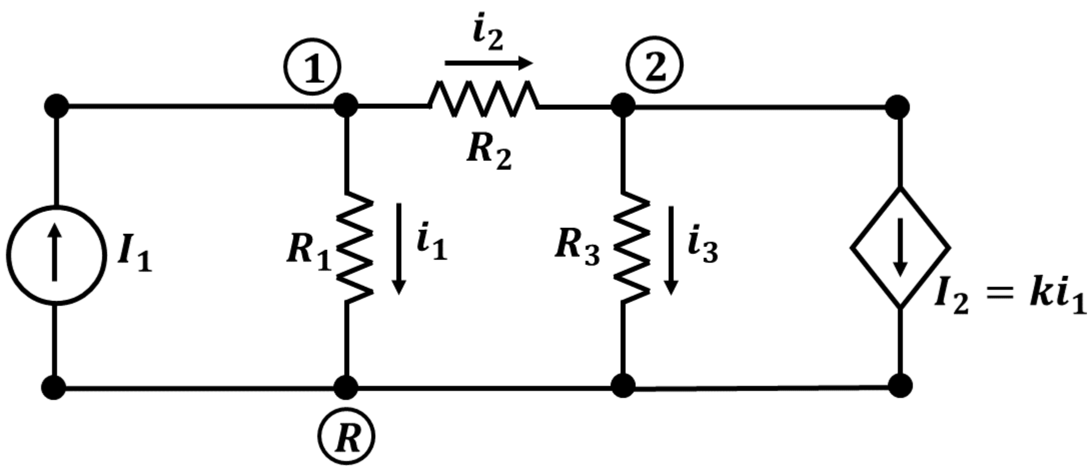
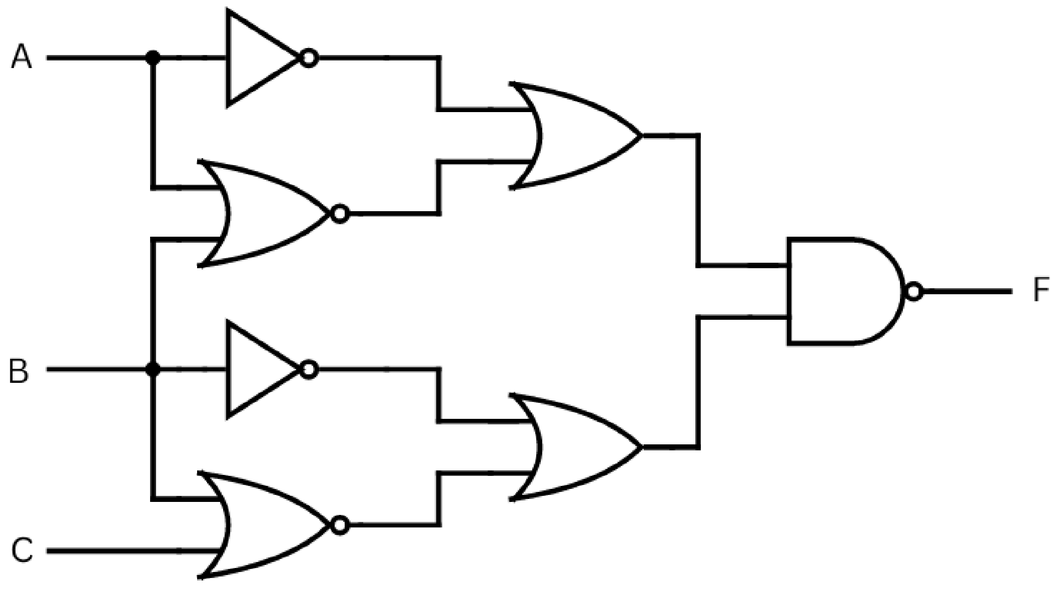


Figure 1: Electric circuit with the independent and dependent current source.

2. Consider the logic circuit shown below in Figure 2.



   (a) Express $F(A, B, C)$ is the Standard SOP form.
   (b) Express $F(A, B, C)$ is the Standard POS form.